MODULE 05 · DOMAIN 1 - SNOWFLAKE AI DATA CLOUD FEATURES AND ARCHITECTURE

Interfaces, drivers, connectors and Git

Many doors into one engine. The exam tests which door, and which one cannot open.

DOMAIN 1 of 5 WEIGHT 31% MODULE 5 of 31

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All of these are clients. The engine is identical. What differs is **what each client can ask for**.

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Snowsight, the web interface

- **Snowsight is the only supported web UI.** The Classic Console was deprecated back in 2023.
- SQL editing, object browsing, dashboards, monitoring, cost management, admin, and the **Query Profile**.
- **Universal Search** across objects, and the Marketplace, are both reached from here.
- **PUT and GET cannot be run from the web UI.** That restriction has its own slide shortly.

> ON THE EXAM
>
> If an option mentions the Classic Console, it is describing an interface that no longer exists.

Worksheets became Workspaces

- **Workspaces replaced Worksheets on 20 April 2026.** Files and folders, sharing, and Git integration.
- **Removal of Legacy Worksheets and Dashboards began 22 June 2026.** Reader accounts follow on 20 September.
- **No native replacement for Dashboards.** A Dashboard-to-Streamlit conversion tool, or third-party BI.
- First use auto-creates a **personal database**, which cannot hold ordinary tables or views.
- The study guide is dated **19 January 2026**, before this change. The exam may still say 'Worksheets'.

> ON THE EXAM
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> Know both names. No question should hinge on it, because the exam predates the rename.

SnowSQL or Snowflake CLI?

SNOWSQL - LEGACY

snowsql

- SQL, DDL, DML, loading and unloading
- Interactive shell or batch with -f
- Variables use &var syntax, now deprecated
- No new features will be added

SNOWFLAKE CLI - CURRENT

snow

- Everything SnowSQL does, plus more
- Streamlit, Snowpark, Container Services, Native Apps, dbt
- Open-source and developer-centric
- All new features land here

THE TIEBREAKER

Snowflake's own docs call SnowSQL legacy and recommend transitioning. If a task involves deploying anything, it is Snowflake CLI.

SnowSQL, the details worth keeping

- Runs as an **interactive shell**, or in batch mode via stdin or the **-f** option.
- Internal SnowSQL commands all begin with an **exclamation mark**, such as !exit and !source.
- Built *using* the Snowflake Connector for Python, though the connector is **not a prerequisite** to install it.
- Support runs by minor release. 1.2.x ended December 2025, 1.3.x ended May 2026, 1.4.x runs to May 2027.
- From version 1.3.0, **automatic upgrades are disabled**. You reinstall manually.

> ON THE EXAM
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> SnowSQL still works and is still supported. Legacy means no new features, not switched off.

PUT and GET: the client-only commands

- **Neither can be executed from the Worksheets page in either web interface.** This is documented explicitly.
- Use **SnowSQL, Snowflake CLI, or a driver** instead. Those can reach your local file system.
- Snowsight separately offers a **file-upload UI** for named internal stages. That is not the same as the PUT command.
- An EXECUTION_STATUS of success in QUERY_HISTORY **does not mean the files actually transferred**.

> ON THE EXAM
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> The reason is obvious once stated: a browser tab cannot be trusted with arbitrary access to your local disk.

Drivers, connectors and the SQL API

WHAT	WHEN IT IS THE ANSWER
JDBC and ODBC	Native drivers. Third-party BI and ETL tools, and Java or C-based applications.
Python, Node.js, Go, .NET, PHP	Programmatic interfaces for application code in those languages.
Spark connector	Reading and writing Snowflake from an existing Spark cluster.
Kafka connector	Streaming topics into Snowflake tables. Covered properly in module 15.
SQL API	A REST API. No driver to install, so it suits serverless functions and lightweight automation.
SnowCD	Not a connection. A diagnostic tool for troubleshooting connectivity.

ON THE EXAM

All of these are Standard Edition. Connectivity is never an edition question.

Git integration: three objects, in order

01

SECRET

TYPE = password. Username plus a personal access token.

02

API INTEGRATION

API_PROVIDER = git_https_api, with allowed prefixes and allowed secrets.

03

GIT REPOSITORY

API_INTEGRATION, ORIGIN, and optionally GIT_CREDENTIALS.

ON THE EXAM

The API_PROVIDER value git_https_api is the detail questions quote. Learn it as written.

What a Git repository actually is in Snowflake

- **A special kind of stage.** Once you see that, everything else about it follows.
- Paths are **@repo/branches/name**, **@repo/tags/name**, and **@repo/commits/hash**.
- **ALTER GIT REPOSITORY … FETCH** pulls a full clone of every branch, tag and commit, and prunes what was deleted upstream.
- **EXECUTE IMMEDIATE FROM @repo/branches/main/setup.sql** runs SQL straight out of the repository.
- SHOW GIT REPOSITORIES, DESCRIBE GIT REPOSITORY, SHOW GIT BRANCHES. An empty repository is not supported.

> ON THE EXAM
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> EXECUTE IMMEDIATE FROM is the payoff. Version-controlled SQL, deployed by running one statement.

Which door for which job

THE JOB	THE CLIENT
Explore data, read a Query Profile, manage the account	Snowsight
Upload or download files with PUT or GET	SnowSQL, Snowflake CLI, or a driver
Deploy a Streamlit app, Native App or Snowpark object from CI	Snowflake CLI
Connect a BI or ETL tool	JDBC or ODBC
Query from a serverless function with nothing installed	SQL API
Run version-controlled SQL from a repository	Git repository plus EXECUTE IMMEDIATE FROM
Diagnose why a connection is failing	SnowCD

ON THE EXAM

Seven rows that answer most of this module. The middle two are where candidates lose marks.

> A platform team is building a deployment pipeline in GitHub Actions. Each run must create or replace several Snowpark stored procedures, deploy a Streamlit in Snowflake application, and then run a set of SQL migration scripts.
>
> Their existing scripts use SnowSQL with the -f option. An engineer proposes keeping SnowSQL to avoid a migration.

What should the team use, and why?

A. Keep SnowSQL. It performs all DDL and DML operations, which covers everything described.

B. Snowflake CLI, because SnowSQL cannot deploy Streamlit apps or Snowpark objects.

C. The SQL API, because command-line clients cannot be used inside GitHub Actions.

D. Snowsight Workspaces with a Git-synced workspace, because deployment must originate in Snowflake.

◆ PAUSE AND DECIDE

What should the team use, and why?

A. Keep SnowSQL. It does all DDL and DML.
True as far as it goes, and deploying a Streamlit app is not a DDL statement. SnowSQL has no command for it.

B. Snowflake CLI. SnowSQL cannot deploy those.
Application deployment is exactly what Snowflake CLI adds, and it also runs the SQL migrations SnowSQL was doing.

C. The SQL API, since CLIs cannot run in Actions.
CLIs run perfectly well in GitHub Actions. The premise is invented.

D. A Git-synced workspace in Snowsight.
Workspaces is an interactive editor. It is not a headless CI runner, and deployment does not have to originate in Snowflake.

THE SIGNAL

A is the trap because its stated reason is true. SnowSQL does do all DDL and DML. The task needed something that is neither.

> An analyst has a 40 MB CSV file on their laptop. They want it in a Snowflake table. Working in Snowsight, they open a SQL editor and run PUT file://~/data/sales.csv @%sales_staging.
>
> The statement fails. They check QUERY_HISTORY and are confused by what they find there.

What is happening?

A PUT cannot be executed from the web interface. It needs SnowSQL, the Snowflake CLI, or a driver.

B The table stage syntax is wrong. PUT requires a named internal stage, not @%table.

C The file exceeds the 25 MB limit for PUT operations and must be split first.

D PUT requires the ACCOUNTADMIN role, which the analyst does not hold.

◆ PAUSE AND DECIDE

What is happening?

A PUT cannot run from the web interface.

Documented explicitly. A browser cannot reach the local file system, so PUT needs a client that can.

B @%table syntax is invalid for PUT.

Table stages are a valid PUT target. The syntax in the scenario is correct.

C Exceeds a 25 MB limit.

Invented. There is no such per-file limit on PUT.

D PUT requires ACCOUNTADMIN.

It requires privileges on the stage, not an elevated system role.

THE SIGNAL

The Query History confusion is the second half of the trap: for PUT and GET, EXECUTION_STATUS success does not mean the file moved.

> A team keeps all its Snowflake object definitions in a private GitHub repository. They want Snowflake itself to run those SQL files, so that a deployment is a single statement rather than a script that pipes files through a client.
>
> They have created a secret holding a personal access token, and a Git repository object referencing it.

What else is required, and how do they run the files?

A Nothing else. Run COPY INTO from the repository stage to load and execute the files.

B A notification integration, then a task that reads the repository on a schedule.

C A storage integration pointing at GitHub, then run the files with EXECUTE IMMEDIATE FROM.

D An API integration with API_PROVIDER = git_https_api, then EXECUTE IMMEDIATE FROM the repository path.

◆ PAUSE AND DECIDE

What else is required, and how do they run the files?

A Nothing else. Use COPY INTO.

COPY INTO loads file contents into table rows. It never executes SQL, and the integration is mandatory.

B A notification integration and a scheduled task.

Notification integrations carry cloud messaging events. Nothing here delivers or executes repository files.

C A storage integration to GitHub.

Storage integrations address cloud object storage such as S3 or Blob. GitHub is reached over an HTTPS API.

D API integration with git_https_api, then EXECUTE IMMEDIATE FROM.

The Git repository object requires an API integration of that provider, and EXECUTE IMMEDIATE FROM runs SQL from the clone.

THE SIGNAL

Four integration types exist and they are not interchangeable. Storage for buckets, API for HTTPS endpoints, notification for messaging, catalog for Iceberg.

> An organisation runs a small serverless function that fires a handful of queries after each nightly load, then writes a summary to a monitoring system. The function runs in a constrained runtime where installing and maintaining native libraries is awkward.
>
> The team also has an existing Spark cluster performing heavy transformations against the same data.

Which combination is most appropriate?

A ODBC for the serverless function, and the Spark connector for the cluster.

B The SQL API for both, since it avoids installing anything anywhere.

C The SQL API for the serverless function, and the Spark connector for the cluster.

D SnowSQL for the serverless function, and JDBC for the cluster.

◆ PAUSE AND DECIDE

Which combination is most appropriate?

A ODBC plus the Spark connector.

Right for Spark, wrong for the function. ODBC is a native driver, which is exactly the install the scenario wants to avoid.

B SQL API for both.

A REST call per statement is wrong for heavy distributed transformations. The Spark connector pushes work down and moves data in parallel.

C SQL API plus the Spark connector.

No driver for the constrained runtime, and the purpose-built connector for the cluster that already exists.

D SnowSQL plus JDBC.

SnowSQL is an interactive client, not a library to call from function code. JDBC would work for Spark but is not the purpose-built option.

THE SIGNAL

Two different constraints in one scenario. 'Installing native libraries is awkward' and 'existing Spark cluster' each point at a different answer.

What to carry into the next module

- **SnowSQL is legacy.** Snowflake CLI gets all new features and handles Streamlit, Snowpark, Native Apps and dbt.
- **PUT and GET cannot run from the web UI.** Use a client. Success in QUERY_HISTORY does not mean the file moved.
- **Git needs three objects:** secret, API integration with git_https_api, and the Git repository. Run files with EXECUTE IMMEDIATE FROM.
- **The SQL API is the no-driver option.** Serverless functions and lightweight automation.
- **Snowsight is the only web UI.** Classic Console went in 2023, Worksheets became Workspaces in April 2026.

> ON THE EXAM
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> Next: Cortex, Snowpark, Streamlit, Notebooks and ML. The biggest single block of new material on this exam.